Fiscal Year:	FY 2019	**Task Last Updated:**	FY 09/05/2020
PI Name:	Kronenberg, Amy Sc.D.		
Project Title:	GCR Simulator Validation Studies with Human and Mouse Models		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-8099	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-RADIATION. Appendix D: Ground-Based Studies in Space Radiobiology
Start Date:	02/04/2016	**End Date:**	09/30/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	6
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Extended to 9/30/2019 per PI and approval by NASA LaRC (Ed., 12/19/18)		
Key Personnel Changes/Previous PI:			
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Grant/Contract No.:	NNJ16HP23I
Performance Goal No.:	
Performance Goal Text:	
Task Description:	This project addresses NASA's request to help design and validate its planned galactic cosmic radiation (GCR) simulator. Space radiation has two major components: GCR, which is comprised of highly energetic ions from protons through uranium, and solar particle events, which are primarily moderate energy protons. Most currently funded NASA ground-based studies in radiobiology examine one highly energetic ion at a time because of facility limitations. While this approach provides insight into the biological effect of each particle, insight into potential interactions from particles of different energies or charges is missed and the deep space environment is not accurately copied. Recent improvements in facility capability allow mixed beam exposures to more closely approximate the GCR, but questions remain about the final composition of the beams to be included. We will build upon our prior work on single particle-induced mutations in cultured cells and mice (cancer-relevant endpoints). Work with cultured cells will be performed because it will provide data more quickly, aid with cross-species extrapolation of risks, and inform the mouse work. We will first measure the effects of acute exposures to alpha particles (charged helium nuclei) because basic information about this GCR component is lacking. Next, we will measure the effect of acute exposures to a mixture of protons, alpha particles, and silicon ions to mimic the effect of a simple GCR simulator model, followed by acute exposures to the final GCR mixture that is chosen (informed in part from our first two experiments). Finally, we will determine the effects of a protracted exposure to the GCR simulator, which will reflect more closely exposures that will occur in the space environment. Our results will help NASA to both design and validate the GCR simulator.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The purpose of this project is to provide input to NASA's risk model for cancer regarding exposure to accelerated charged particle radiations that simulate those found in space. The work is designed to address cancer development processes at relatively low doses and fluences that are space-relevant. The project specifically uses different types of charged particle radiations alone or in combination to determine whether it is possible to predict the effects of multiple ions from the effects measured in studies using single light or heavy charged particle beams. This work is of relevance to naturally occurring Earth-based exposures to densely ionizing radiations such as radon. It will also inform our understanding of the mechanisms underlying potential risks to normal tissues associated with the implementation of charged particle radiation therapy.
Task Progress:	Progress in the final period of this project included the completion of studies with 3 sequential ion beams rapidly delivered to mouse kidney epithelial cells in culture and a cohort of B6D2F1 mice. The results showed simple additivity for the genotoxicity studies (cell death, autosomal mutation) with cultured kidney cells. Mutagenesis studies in the mouse model produced less than additive results at the highest dose point. Subsequent studies were performed using both the cultured kidney cells and a separate cohort of B6D2F1 mice exposed to 6 sequential ion beams to more closely resemble the complex GCR environment. The results were similar to the results obtained with 3 sequential ion beams. In addition, supplemental behavioral and cognitive studies were performed with the mice exposed to 3 beams or to 6 beams. There was considerable variation in the results between the two studies, but the object recognition studies showed similar effects of exposures to complex charged particle beams of importance for space flight. Human lymphoid cell studies included exposures with He ions alone to round out an extensive set of single ion studies funded by NASA over many years. The results indicate that high energy He ions have similar toxicity and mutation induction to equal doses of energetic protons, but that they are 4x more effective than protons on a per particle basis. Additional studies examined the effect of combined exposures to 3 ions delivered rapidly. The results indicate that the toxicity of the combined exposure could be predicted from the single ion results, as could the mutant frequency data. Dose-rate studies with multiple ions were performed using two approaches: continuous low dose-rate or split dose with moderate dose-rate. The results indicate reduced cytotoxicity for low dose-rate exposures but little or no sparing for mutant frequency. Molecular analysis using whole genome SNP (single-nucleotide polymorphism) analyses was initiated on a large number of murine kidney epithelial samples from some single and multi-beam exposures and for human lymphoid cells. The results are pending.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
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